XINRONG ZHU

San Ramon, CA | 510-556-6208 | xinrongzhu.eng@gmail.com | linkedin.com/in/xinrongzhu/ | Portfolio | US Permanent Resident

EDUCATION

Cornell University, College of Engineering | GPA: 3.92/4.0 Ithaca, NY
Bachelor of Science in Electrical and Computer Engineering, Minor in Robotics Expected May 2027
Relevant Coursework: Circuits, Digital Logic and Computer Organization, Mechatronics, Embedded Systems, Microcontrollers, Embedded OS, Robotics, Nanoscience and Nanotechnology (Teaching Assistant)

TECHNICAL EXPERIENCE

Electrical Engineering Intern, DEKA Research and Development, Manchester, NH Jun 2026 - Aug 2026
- Developed electrical hardware and Python firmware for AirXRP, an autonomous educational quadrotor selected as the official starter platform for the STEM-ED Aerospace Robotics Competition, with 35 kits ordered for student use
- Designed a custom KiCad PCB shield for an RP2350-based XRP controller, interfacing with a flight controller and peripherals through UART, SPI, and I2C
- Developed Python libraries for drone control, autonomous navigation, and sensor integration. (github)
- Diagnosed recurring motor failures during high-payload flight testing; optimized PID and control-loop parameters to achieve reliable flight with a 2 lb payload for 10 minutes

Electrical Engineering Undergraduate Researcher, Space Systems Design Studio, Ithaca, NY Sep 2026 – Present
- Design a custom PCB schematic and layout in Altium Designer for a handheld ChipSat detector using an ATmega328P IC, Lora transmitter, external display, SPI programming interface, and power regulation
- Perform SMD and THT PCB assembly, board bring-up, electrical testing, and antenna impedance tuning to validate RF communication and troubleshoot hardware

Undergraduate Researcher, Collective Embodied Intelligence Lab, Ithaca, NY Jan 2026 – Present
- Develop a robotic environmental DNA (eDNA) sampling system from initial concept to a working prototype, including component selection, mechanical design, electronics integration, and system testing, featured and presented in the NERC 2026 Special Exhibition section, received Cornell's Fall 2026 Dean Archer Undergraduate Research Award
- Design and wire the sampler's power and control circuitry, programming an Arduino with motors, pumps, servos, sensors, and linear actuators using motor drivers and transistor switching
- Develop an ESP32 based wireless water-quality monitoring system, integrating eDNA sampler hardware with a custom chlorophyll fluorescence sensor and Wi-Fi communication for real-time lake deployment

PROJECTS

Mixed Signal Birdsong Synthesizer Ithaca, NY
- Built an RP2350-based birdsong synthesizer using a potentiometer, MCP4822 DAC, ADC inputs, and SPI; implemented direct digital synthesis in C and validated analog waveforms with an oscilloscope

Hello Attack - Interactive Greeting Robot (github) Ithaca, NY
- Developed embedded C firmware on an ARM Cortex-M0+ microcontroller, integrating ultrasonic sensing, servos, and LEDs; implemented low-level PWM drivers from scratch using timer registers and verified timing with an oscilloscope

LEADERSHIP EXPERIENCE

Team Lead, Cornell Cup Robotics, Ithaca, NY Jan 2025 - Present
- Lead and mentor a 20+ member team, overseeing technical projects and recruitment while providing hands-on training in machine-shop fabrication, CAD, 3D printing, technical documentation, and laboratory safety
- Develop and validate mechanical assemblies for robotic systems using Fusion 360, 3D printing, and machine-shop fabrication, applying DFM/DFA principles to improve part design, assembly efficiency, and mechanical reliability
- Present projects to students, governors, mentors, at FIRST Robotics Governor's Cup and Worlds Championships

SKILLS

Electrical: Power, PCB Design (Altium, KiCad), Motor Drivers, Controls, RF Testing, Oscilloscopes, Soldering (SMD/THT)
Embedded/Software: Arduino, ESP32, Raspberry Pi, ARM, UART, SPI, I2C, PWM, C/C++, Python, Java, MATLAB, Git
Mechanical: Fusion 360, 3D Printing, CNC Machining, Laser Cutting
Work Authorization: Authorized to work for any US employer (no sponsorship required)